

A federated and explainable approach for insider threat detection in IoT

Mohammad Amiri-Zarandi^a, Hadis Karimipour^b, Rozita A. Dara^{a,*}

^a School of Computer Science, University of Guelph, Guelph, Canada

^b Department of Electrical and Software Engineering, University of Calgary, AB, Canada

ARTICLE INFO

Keywords:

Insider threat detection
Internet of things
Trustworthy learning
Privacy
Security
Federated learning
Explainable artificial intelligence
Interpretable artificial intelligence
Anomaly detection

ABSTRACT

An insider threat is a malicious action launched by authorized personnel inside the organization. Since insider actions may only leave a small digital footprint in the system, it is considered a major cybersecurity challenge in different applications. Along with the rapid growth of the Internet of Things (IoT) and the extensive attack surface in this technology, many concerns have been raised regarding the potential insider threats in IoT environments. Several studies have been conducted on Machine Learning (ML)-based insider threat detection solutions which are focused on the models' performance while the trustability of these models is neglected. Trustworthy Learning refers to a new trend in ML that focuses on ways to ensure that the data collection and data analysis procedures in ML techniques follow ethical applications and are trustable to human users. This approach enforces the acceptance and successful adoption of ML-based solutions. This study aims to propose an improved trustworthy insider threat detection method that ensures two of the trustworthy learning requirements: Privacy and Explainability. The proposed solution protects the privacy of the utilized data and is capable of explaining why certain behaviors are detected as a threat. The proposed solution also leverages data collaboration between different data owners to increase the volume of data used in the training process and enhance the performance of the ML model. Experimental results show the proposed solution outperforms the learning models trained by individual data holders.

1. Introduction

Today's highly interconnected world is made up of billions of smart Internet of Things (IoT) devices that collect, store, and analyze data regarding different aspects of our daily lives. IoT technology enables smarter and faster control of our daily actions. [1,2,3] Despite the benefits that IoT brings to our lives, it increases the vulnerability of the security frameworks to cyber threats. [4,5,6,7] The attack surface in IoT is significantly expanded as insiders and intruders can launch cyber attacks by accessing IoT devices remotely. To secure IoT against adversaries, cybersecurity experts not only must try to eliminate intrusion attacks by securing the devices, network, and software. [8,9] but they also should take measures to mitigate insider threats. An insider threat happens when a user or a business partner that has the authorization to access the resources such as the system and data aims to misuse the access capability to intentionally or unintentionally cause a threat in the system. [10] The security of IoT can be seriously compromised by insider threats. It has been reported by the IBM Security Intelligence Index [2] that 60% of all attacks are launched from the inside. According to Gurucul's

* Corresponding author.

E-mail address: drozita@uoguelph.ca (R.A. Dara).

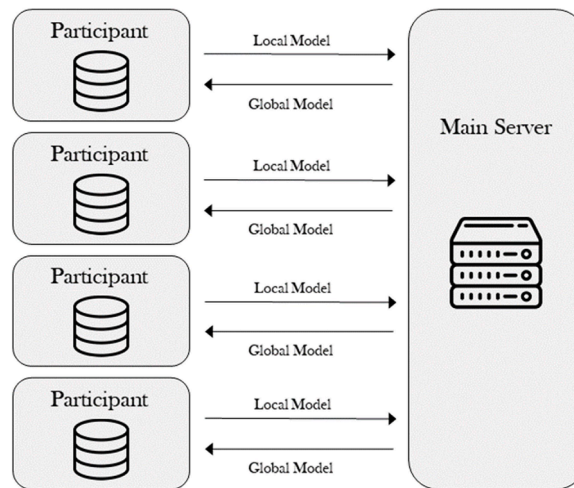


Fig. 1. Federated Learning Architecture.

2020 Insider Threat Report [11], 68% of the cybersecurity experts responded that their organizations are vulnerable to insider threats at medium or high levels while merely 5% believed they are not vulnerable at all. In spite of the high importance of insider threat detection in IoT, a few studies have been conducted on this aspect of IoT security. [12]

Previous research has proved that Machine Learning (ML) is an effective approach to insider threat detection. [13] ML-based models can utilize historical data about previous actions in the system and classify a new behavior as benign or malicious. The ML-based intrusion detection systems analyze different data about users including demographic information, position and responsibility of the users, and also behavior patterns of the users in the system to detect suspicious activities. For example, a user logs on to a device after hours and uploads data to an untrusted website, or a user tries to log in to other users' devices and run executive files. Analysis of users' information enables security officers to discover the early signs of malicious actions and take proper actions. When the insider threat detection system finds a potential threat, it sends an alert to security officers for further investigation. This can lead to some decisions such as preventing the action, blocking the device, or suspending the user.

One significant concern in ML-based insider threat detection systems is the trustability of these systems. [14] This is due to the fact that the analyzed data contain users' personal information which is sensitive and should be protected. Furthermore, the explainability of the ML-based model helps cybersecurity experts detect the causing factors of insider threats and take measures to mitigate the risks of these threats in the future. Moreover, due to the potential impact that the detected threats will have on the related users' jobs and status, the detection mechanisms should be designed in an explainable manner. This enables the security officers to illustrate why a particular action is considered a threat. To address privacy concerns in ML methods, one solution is to follow the Trustworthy Learning approach. Trustworthy Learning refers to a new trend in ML that focuses on ways to ensure that the data collection and data analysis procedures are trustable. [15] Since ML approaches have shown the capability to demonstrate and even surpass human intelligence, this approach aims to address the concerns that have been raised about the ethical application and trustworthiness of ML models. [16] Trustworthiness becomes a bigger issue when human data is used in the learning process, thus the privacy of sensitive personal information should be protected; or when the resulting predictions can be the basis for the decisions and actions related to human users, therefore the causes for the decisions should be explainable for the users. [15] In this paper, we aim to propose a trustworthy ML approach for intrusion detection in IoT which still is a gap in the literature. Different requirements for Trustworthy Learning have been suggested in previous studies. [16,17,18,19] In this study, we focus on two of these requirements that are common in most Trustworthy Learning requirement categories: privacy, and explainability. Privacy is the characteristic that information is not made available or disclosed to unauthorised individuals, entities, or processes. [20] Data protection mechanisms reduce data privacy concerns and encourage the data holders in the IoT ecosystem to utilize data analysis services. [21,22,23] On the other hand, explainability refers to a property that provides visibility into how ML models make predictions. [24] This characteristic provides insight into the rationale for results generated by ML models and helps understand and interpret these results.

In this study, we propose a threat detection system that analyzes user behavior to discover early signs of malicious behaviors. These behaviors are flagged as suspicious actions for further investigation by cybersecurity experts. The proposed method is explainable and privacy-preserving and thus, addresses two of the major requirements of Trustworthy Learning. To the best of our knowledge, this is the first attempt to propose a privacy-preserving solution for insider threat detection in IoT, and also the first study on providing explainability on this problem.

The contribution of this research can be summarized as follows:

- This research proposes a privacy-preserving approach for insider threat detection in IoT.
- The study suggests an explainable approach that provides an illustration of the most contributing features to insider threats and the role of each feature.

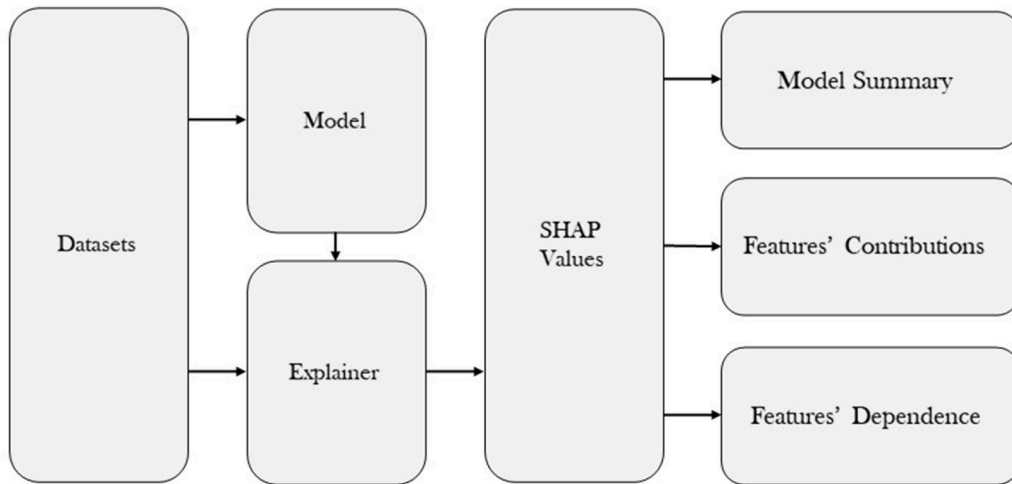


Fig. 2. An overview of the SHAP method.

- Experimental results on a public dataset of insider threats show that the proposed solution is capable of generalizing and detecting insider threats under limited investigation budgets.

The remaining paper is organized as follows: A review of the background and the related works in the literature is provided in Section 2. Section 3 presents the proposed threat detection system. In Section 4 a comprehensive evaluation of the proposed solution is provided. Section 6 discusses the findings and limitations. Finally, the conclusions and future work are provided in Section 5.

2. Background

2.1. Federated learning

Federated Learning is an approach that can be used to address the privacy concern in ML-based solutions. Federated learning first was proposed by Google. [25] This approach involves two fields that are rapidly evolving: Artificial Intelligence (AI) and Information Privacy (IP). Instead of gathering data to train a centralized ML model, the federated learning approach distributes the learning model to devices that have stored data locally. [26,27]. The distributed models are trained on the local data and then are aggregated to build a global ML model. Using this approach the participants do not need to share data with any other parties thus data privacy is preserved. [28] An overview of the federated learning architecture is demonstrated in Fig. 1. The distributed nature of federated learning architecture is well-aligned with IoT architecture. The IoT devices can participate in the federated learning process and collaborate to train an ML model while keeping their data private. This cooperation can be handled by IoT devices that have enough computation power for model training or the IoT gateways that are placed at the edge of the IoT network. [29,30]

2.2. Explainable learning

The other Trustworthy property that we consider in this study is explainability. This property enables the ML model to explain the reason for a prediction. An explanation of why an instance is detected as a threat provides insight into the most causing features and enables cybersecurity experts to improve security monitoring practices. Moreover, once an anomalous behavior is detected, the cybersecurity professionals can provide the involved users with the reasons why they are under investigation. In contrast to the black-box concept in neural networks where even the algorithm designers cannot explain why a model arrived at a specific decision, the Explainable AI tries to provide the decisions with reasons that can be understood by humans. [31] The Explainable AI methods can be categorized into two different classes: first, local methods that can explain the reasons for specific predictions, and second, global methods that are capable of summarizing the impact of input features on the model as a whole. [32] LIME (Local Interpretable Model Agnostic Explanations) [33] is a sample of local explainable methods that reflect the behavior of the model around the instance being predicted. This method changes the input data and builds a series of manipulated data containing only some part of the original data. Then by evaluating the performance of the model in the presence or absence of certain features, the method evaluates the impact of each feature in prediction. SHAP (Shapley Additive Explanation) [34] is an instance of global methods. This method uses a game theory approach to explain the predictions of ML models. According to this approach, the feature values are presumed as players that work in a collaborative game, while the payout is the prediction. As a result, Shapely values describe fairly how each feature contributes to the predicted result. [32] An overview of using the SHAP method in explainable ML is shown in Fig. 2. Using this method, first, the model is trained over the training dataset. Then the trained model and the training data will be shared with the SHAP explainer which calculates the SHAP values for different features and provides some explanations such as the average influence of all

Table 1
Comparison of prior studies.

Paper	Method	Database
[1]	Distance-Based Analysis	Synthetic Data Based on CERT Dataset
[2]	LSTM	CERT v6.2
[3]	LSTM, GRU	CERT v4.2
[4]	LSTM Autoencoder	CERT v4.2
[5]	Autoencoder	CERT v6.2
[6]	LSTM	CERT v6.2
[7]	Log2vec	CERT v6.2

features in predicted labels, features' contribution for every single data sample, and the dependence among the involving features.

3. Related work

Insider threat detection systems are crucial to prevent malicious activities caused by users that are authorized to access the resources in an environment. These systems collect data about previous activities to find the pattern of benign and malicious behaviors. Lu and Wang [35] leverage deep learning to discover insider threats in an organization. As part of the proposed methodology, the sequence of user activities is treated as a time series and a Long Short-Term Memory (LSTM) model is used to identify patterns indicating normal behavior to benign activities and malicious actions. Al-Mhiqani et al. [36] used a similar approach to analyze the action sequences using deep learning, while in this study Gated Recurrent Unit (GRU) is used for time series analysis. According to this paper, similar to LSTM, GRU can capture long-term temporal dependencies on the sequence of user actions, but the GRU model is more efficient in terms of computational resources needed and training time. Sharma et al. [37] proposed an anomaly detection mechanism for insider threat detection using an LSTM-based Autoencoder. In this paper, it has been recommended to analyze the users' actions within respective users' sessions which is the time period between a user's sign-in and sign-out. Liu et al. [38] also presented a method based on autoencoders for user behavior analysis using anomaly detection. This study uses an ensemble of autoencoders that independently learn from training data and the final result is generated using an algorithm designed to aggregate all anomaly scores to a single value. Kim et al. [39] provided a survey of insider threat detection approaches from IoT perspective. In this study, different data sources are analyzed based on the characteristics and the structure of IoT. The authors demonstrated that the datasets that include data from the network and application layers of the network are better choices than the ones that are created based on the perceptual layer data. Khan et al. [10] proposed an insider threat detection approach that is specifically designed for IoT. This approach is lightweight and decreases the needed computation resources. Matterer and LeJeune [40] designed an ensemble of LSTM models that leverage the original dataset along with peer group metadata for detecting suspicious behaviors. The authors defined the peer groups based on the users' roles, units, departments, and teams. Liu et al. [41] suggested to use log2vec, a heterogeneous graph embedding-based algorithm to find insider threats. This algorithm builds a graph based on system logs and then uses this graph to convert each log entry into a low-dimension vector. Finally, these vectors are fed into the log2vec algorithm to distinguish normal logs and malicious ones. A summary of these studies is presented in Table 1.

In this study, we aim to design a trustworthy ML-based solution for insider threat detection. A major requirement in Trustworthy Learning is privacy. Because insider threat monitoring systems need to utilize user data, these systems raise concerns regarding the users' privacy, thus it is hard to convince data holders to share data for training ML models. [42] While certain approaches may incorporate anonymized data for behavioral analysis, previous research has demonstrated the inadequacy of anonymization in preserving privacy. These studies have revealed that, on occasion, the amalgamation of de-identified data with external data sources enables attackers to reestablish the link between anonymized data and the underlying sensitive personal information. [8] To provide more robust privacy protection, recently Federated Learning has been utilized in some security applications such as intrusion detection and threat intelligence [43,44], to address the issues related to privacy. Rey et al. [45] presented a framework for malware detection in IoT using federated learning. In this study, two approaches were utilized to design the ML model: first, a supervised learning approach using neural networks, and second, an unsupervised learning approach using Autoencoders. Preuveneers et al. [46] designed a blockchain-based federated learning method in which the updates on the learning model are stored on distributed ledgers. This method provides a distributed approach both in the learning and storage processes. Fan et al. [47] proposed an intrusion detection framework for 5 G IoT. The proposed framework leverages federated learning along with transfer learning for intrusion detection. The role of transfer learning in this study is to train a model on a dataset and transfer the trained model to a different space to be used as the initialized model in federated learning.

Similar to federated learning, Explainable AI is a fast-evolving topic in AI which in many studies is listed as a necessary requirement for Trustworthy Learning. [16,17,18,19] Kim and Routledge [48] discussed the impact of explainability on user privacy and also its role in decision-making processes. Kopp et al. [49] use random forests to extract some rules explaining the reason for a prediction in the learning process. Antwarg et al. [50] proposed a method to explain the outliers detected by unsupervised models that utilize Autoencoders. This method provides a comprehensive explanation of the relation between the features with high anomaly error and the features that are most causing this anomaly error.

In the literature, the insider threat detection systems that address trustworthy learning concerns are still a gap. This paper aims to propose an ML-based method for insider threat detection in IoT focusing on two of the Trustworthy Learning requirements which are common in most Trustworthy Learning requirement lists: privacy, and explainability. The proposed method is privacy-preserving and

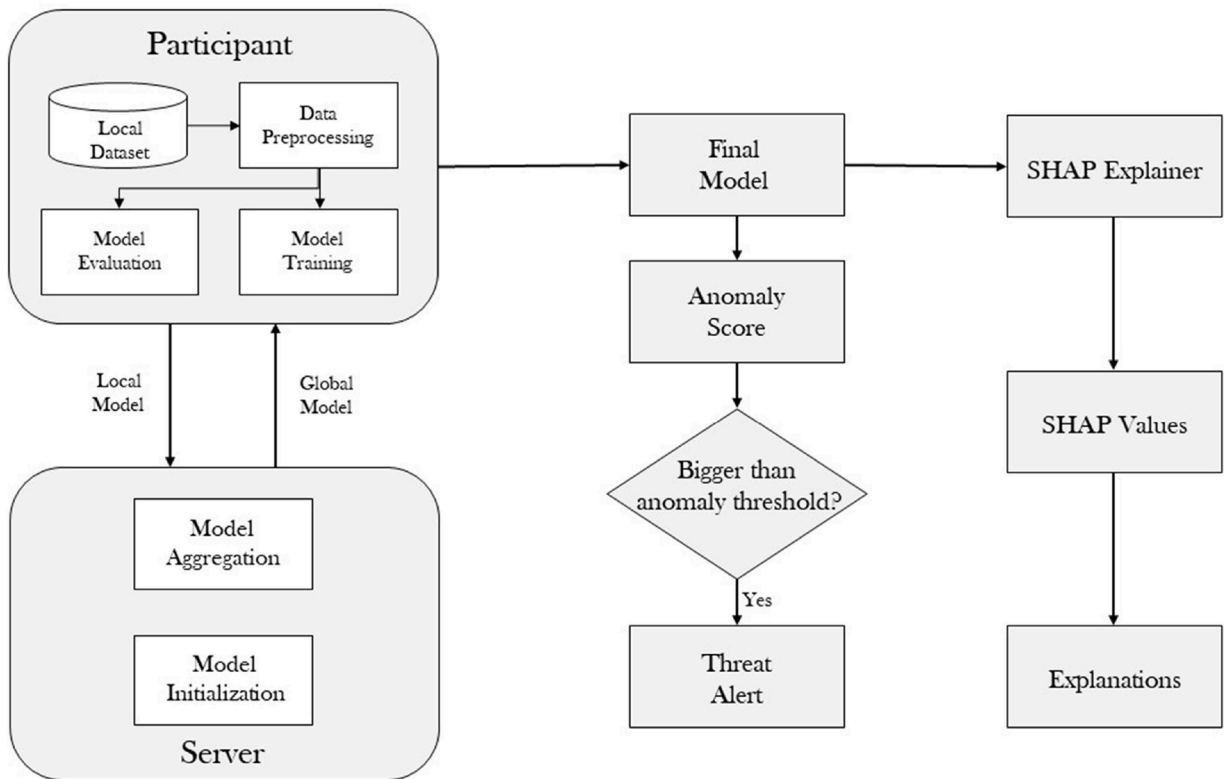


Fig. 3. An overview of the proposed threat detection system.

also is able to explain the causing factors in the discovered threats.

4. Proposed approach

In this section, we propose a trustworthy insider threat detection model that improves the available ML-based solutions in terms of privacy and explainability. This model analyzes user data to detect possible threats while ensuring the privacy of the users and describes the reasons for the predicted threats by leveraging SHAP, a model-agnostic explainable method. An overview of the proposed threat detection system is depicted in Fig. 3. Using the federated learning approach an ML model is trained over local user data in collaboration among some participants. A participant can be an organization that manages an IoT network and has access to data regarding the users that work with the devices in the network. These data can include users' personal information (e.g. personal identification, demographic data, and psychological data) and historical records of users' activities in the system. Since the participants do not need to share the user data with others, the privacy of the data is preserved. The final model generated through the federated learning process will be utilized by each independent participant to discover the threats in the systems.

Moreover, this model will be analyzed using SHAP to evaluate the more affecting features in detected insider threats. The SHAP model provides an explanation of the involvement of every feature in every single discovered insider threat as well as an overall overview of the most impacting features in potential insider threats. Section 4.2 provides more details on the utilized federated learning algorithms, and Section 4.3 describes the used SHAP algorithm in more detail.

Although insider threat detection systems have access to large data regarding users' activities, usually these data contain a small number of malicious actions. These actions might be a few instances over a decade which means the data might include less than 0.1% of malicious actions. [51] Detecting these few malicious actions among the extensive normal activities is a complex task that usually needs to uncover nonlinear patterns in the users' activities. Due to the limited available malicious data samples, the anomaly detection approach is suitable for detecting insider threats. [38] Anomaly detection is an approach that tries to detect data samples that significantly deviate from the majority of data samples. [52,53,54] Through this approach, normal instances are considered as a baseline and any deviation from these normal samples can be treated as an outlier or anomaly. It has been shown that anomaly detection methods provide a promising performance when most instances in the data can be treated as normal. [55] In this study, we use a deep autoencoder architecture as the base learning model due to its capability to detect nonlinear patterns and its ability in anomaly detection. [56,8] An autoencoder model can encode a normal user behavior instance to a compressed representation, called latent representation, and decode it to the original instance with minimum error. Therefore, any malicious actions operated by insider adversaries lead to a significant error value and can be detected as a threat.

Table 2
Notations.

Acronym	Description
N	The Number of Participants
P_i	The i th Participant
D_i	The i th Local Dataset
S	The Main Server
M_i^r	The local model generated by nod i in the round r
M_G^r	The global model generated in the round r
w_i^r	The weights for the local model generated by nod i in the round r
w_G^r	The weights for the global model generated in the round r
η	Learning Rate
n_i	The volume of training data in D_i
AE	Autoencoder model
X	Input sample of autoencoder
X'	The output of the autoencoder (Reconstructed sample)
x_i	The i th feature in the input sample
x'_i	The i th feature in the reconstructed sample
RE	A list of reconstruction error values
AFL	Anomalous feature list

In the training phase, the autoencoder minimizes the loss function as follows:

$$\phi, \psi = \arg \min_{\phi, \psi} L(X, X') \quad (1)$$

Where ϕ and ψ are encoder and decode respectively. The encoder converts the input vector (x) to the latent representation vector (Z). The output of the i th encoding layer (z^i) can be inferred from the following formula:

$$z^i = \sigma^i(W^i x + b^i) \quad i = 1, 2, \dots, n \quad (2)$$

In this formula, n is the number of layers in the encoder, and σ^i is the activation function in the i th encoding layer. Similarly the output of the j th decoding layer (x'^j) can be calculated as follows:

$$x'^j = \sigma^j(W^j Z + b^j) \quad j = 1, 2, \dots, n \quad (3)$$

In the proposed architecture all participants train a copy of the same autoencoder on their local data and participate in a federated learning process to generate a global autoencoder.

4.1. Problem definition

We assume that there are some N participants in the system each one having access to a different dataset of users' behavior for training. The participants are designated as $\{P_1, P_2, \dots, P_N\}$ in this study each associated with a local training dataset denoted by $\{D_1, D_2, \dots, D_N\}$. Local datasets are considered to have the same features but may differ in terms of volume and distribution. These local datasets include data regarding user behavior activity and we presume the participants are not willing to disclose these data to other parties. The utilized autoencoder model consists of n layers in the encoder part of the model and the same number of layers in the decoder. The input vector is denoted as X , and x_i represents the i th feature of the input data while x'_i is the reconstructed value for the i th feature.

The goal of this study is to train a trustworthy threat detection model that uses participants' cooperation to achieve a higher performance than when they work independently. In the federated learning process, we assume a server, denoted by S , is responsible for distributing the learning model to the participants. After each round, this server collects and aggregates the trained models in a global model. This server is presumed that has no access to the local training data, and only the model parameters will be shared with this server.

Table 2 lists the key notations utilized in the proposed approach.

4.2. Privacy-preserving threat detection

To preserve user data privacy, the proposed architecture leverages a federated learning approach in which participants train unsupervised anomaly detection models and collaborate to enhance the model performance by utilizing the federated learning approach while privacy preservation is assured. A participant can be either an IoT device with enough computational and storage resources to undertake the local learning process or it can be a middleware device that connects devices with limited resources to the network (e.g., an edge server).

As is shown in Fig. 3, in the proposed architecture, each participant trains a model on the local data, and then only the model

Algorithm 1

The learning procedure of the proposed approach.

```

Input: The list of participants (P), The local datasets (D)
Output: The global model
3: S Initializes  $M^0$ 
4: for each round  $r = 1, 2, \dots$ :
5:   for each  $P_i$  in  $P$  (in parallel):
6:      $w_i^{r-1} = w_{Global}^{r-1}$ 
7:      $B \leftarrow$  Splitting  $D_i$  into batches size  $B$ 
8:     for each epoch  $e = 1, 2, \dots$ :
9:       for each batch  $b$  in  $B$ :
10:         $w_i^e \leftarrow w_i^{e-1} - \eta \Delta L(w_i^{e-1}, b_i^{e-1})$ 
11:      end for
12:    end for
13:   $w_{Global}^r \leftarrow \frac{\sum_{i=1}^N n_i w_i^r}{\sum_{i=1}^N n_i}$ 
14: end for
end procedure

```

Algorithm 2

Compute the Shapely values for the most anomalous features.

```

Input: Sample (X), Autoencoder model (AE)
Output: Shapely values for the most anomalous features (shapValues)
1:  $X' = AE.predict(X)$ 
2: For each  $x_i$  in  $X$ :
3:    $RE[i] = L(x_i, x'_i)$ 
4: end for
5: AFL = features with high values from RE
6: for each  $f$  in AFL:
7:   explainer = shap. KernelExplainer(AE, X)
8:   shapValues[f] = explainer.shapvalues(X, f)
9: end for
end procedure

```

parameters are shared with a server. This server has no access to the local datasets and merely is responsible for handling model distribution and aggregation. The final model trained in this system is utilized to find anomaly behaviors that should be investigated by the cybersecurity team.

The learning and aggregation process of the proposed system is described in [Algorithm 1](#). In the first step, the server (S) initializes a model and broadcasts this model to selected participants. Each participant (P_i) trains this model on private local data (D_i). The goal of the local training process is to minimize the Loss function (L) and update the local model's weights (w_i^r). Then the server collects the locally trained models from the participants and aggregates them in a global model (M_{Global}) using weighted averaging. In the next stage, the server sends the global model back to the participants to repeat the same procedure. This procedure continues for some rounds and then the participants can individually utilize a copy of the global model for insider threat detection.

4.3. Explainable learning

In this study, we utilize SHAP (Shapley Additive Explanation) [34] to provide an explainable unsupervised anomaly detection for threat detection in IoT. This model provides a global overview of the features' effect on label values and also explains the contribution of the features on each data sample's label. A challenge in explainable unsupervised learning is unlike the supervised models, we have no predicted label to evaluate the contributions of each feature. To tackle this problem, we designed the proposed model based on the method proposed by Antwarg et al. [50] which suggests utilizing reconstructed features as predicted values.

In our proposed model, an autoencoder is responsible for reconstructing the input sample values. The quality of the reconstructed values can be described using an anomaly score (e.g. reconstruction error) which is the error between the input sample and the reconstructed output. The anomaly samples are the ones that have high anomaly score values. To find out why a sample is not created well, we can follow the following steps.

- (1)- Evaluate the reconstructed outputs to find the anomalies.
- (2)- Identify the features with high feature reconstruction errors that cause the anomaly.
- (3)- Compute the Shapely values for the identified features to evaluate the contribution of other features in the anomaly score.

This procedure is described in [Algorithm 2](#). In this algorithm, X indicates an input sample with the feature set $(x_1, x_2, x_3, \dots, x_d)$,

Table 3

The CERT v6.2 files description.

Files	Description
Organization Structure	Information on the departments, units, and users' positions.
Psychometric	The big five psychometric scores for each user.
Logon	Information regarding users' login and log-off along with computer ids and time stamps.
File	Users' file transfer information, file format, and time stamp.
Email	E-mail exchange data including e-mail size, attachment count, internal/external recipient.
HTTP	The URLs visited by users, the computer id, and some keywords selected from the web page's content.
Device	Device connection information for each user, containing the computer ids, the timestamps of connection/disconnection.

Table 4

Different Categories Utilized for Feature Generation. [58]

Feature Category	Feature Types
Statistics of HTTP Webpages	url length, url depth, content length, content #words
Action on HTTP webpage	Visit (act1), download (act2), upload (act3)
Type of HTTP Webpage	soc. net., cloud, job, hacking, info. leak, other
Statistics of Emails	#dest., #attachments, #attachment types, attachment size, #external dest., #bcc dest., size, #words
Email Content	external, external bcc
Action on Email	receive (act1), send (act2)
Statistics of File	Len, depth, #words
Action About File transfer	to usb (act1), from usb (act2), disk (act3)
Action on Files	copy (act1), read (act2), write (act3), delete (act4)
Type of File	compressed, photo, document, text, exe, other
Statistics of USB Connections	connect duration, file tree length
Log in/out Actions	login (act1), logout (act2)
Workstation Type	user's PC, shared PC (pc0), supervisor's PC, other PC
Time	workhour, after workhour, weekend

Table 5

Number of Data Samples in Different Local Datasets.

Local Dataset	Number of Samples
1	93,341
2	6534
3	249,324
4	248,668
5	253,924
6	136,043
7	129,983
8	4902
9	134,182
10	136,361
11	712

and X' is the corresponding reconstructed sample with the reconstructed feature values as $(x'_1, x'_2, x'_3, \dots, x'_d)$. The algorithm first predicts the reconstructed values for the input sample (X) using the autoencoder. Then a list of reconstruction errors (RE) is calculated for all features. In the next step, the features with high reconstruction values are stored in a list (AFL). These features have the most impact on the autoencoder to be unable to reconstruct the input sample and therefore cause the anomaly. [50] In the next step, these features are used like predicted labels, and the contribution of all other features in generating such labels is calculated using Shapely values.

This method sheds light on the features that are more important in threat detection in a cybersecurity system and helps cybersecurity experts get a better understanding of the reason that one specific behavior is detected as suspicious in the system.

5. Evaluation

5.1. Dataset

As is shown in Table 1, the datasets that widely are employed in various studies on insider threat detection are CERT v4.2 [37] and CERT v6.2. [57] CERT v4.2 includes 1000 employees, where 70 are malicious insiders under three threat scenarios. On the other hand, CERT v6.2, which is the newest CERT dataset, depicts a much larger company with 4000 employees, containing only five malicious insiders (five threat scenarios, with only a single malicious user per scenario). This brings the detection task in CERT v6.2 closer to

Table 6

Losses of different local models in the federated learning process.

Participant Round	1	2	3	4	5	6	7	8	9	10	11	Average
1	0.52	0.58	0.62	0.68	0.65	0.67	0.65	0.58	0.66	0.63	0.66	0.63
2	0.53	0.54	0.54	0.55	0.54	0.58	0.58	0.56	0.55	0.55	0.61	0.56
3	0.53	0.54	0.53	0.53	0.53	0.57	0.57	0.55	0.54	0.54	0.59	0.55
4	0.53	0.54	0.53	0.53	0.52	0.56	0.56	0.55	0.54	0.53	0.57	0.54
5	0.52	0.53	0.52	0.52	0.52	0.55	0.56	0.54	0.53	0.53	0.55	0.53
6	0.52	0.53	0.52	0.53	0.52	0.55	0.55	0.54	0.53	0.53	0.55	0.53
7	0.52	0.53	0.52	0.52	0.52	0.55	0.55	0.53	0.54	0.53	0.55	0.53
8	0.52	0.53	0.52	0.53	0.52	0.55	0.55	0.53	0.53	0.53	0.55	0.53
9	0.52	0.53	0.53	0.53	0.53	0.55	0.55	0.53	0.53	0.53	0.55	0.53
10	0.53	0.53	0.52	0.53	0.53	0.56	0.55	0.53	0.53	0.53	0.55	0.53

real-world scenarios while simultaneously introducing greater complexity. Considering these factors, we chose to evaluate the performance of our model using CERT v6.2. Although this dataset was not originally designed for IoT environments, Kin et al. [39] have illustrated how CERT can be utilized for behavior analysis in IoT threat detection systems. The dataset includes organization structure data and user psychometrics information of the employees in a virtual origination. The dataset includes also activity behavior logs including login/out, file transfer, emails, visited webpages, and connected external devices.

A summary of the dataset files is depicted in Table 3. We utilized the feature extraction method suggested by Le et al. [58] to extract features based on the statistics of users' daily behaviors. Using this method 892 different features are extracted each describing one aspect of user behavior each day. Table 4 describes different categories utilized for feature generation. The name of the features is selected based on the information that the feature carries on. As an example, the feature that is named *afterhourfile_n_exef* contains the number of files executed after work hours in a day.

To design the experiments for the federated learning approach, we divided the data based on the user's unit. CERT v6.2 users belong to 11 different work units. We presumed that each unit is an independent local participant that aims to collaborate with other participants in a learning process but is not willing to share its data with others. The number of training samples in different local participants is demonstrated in Table 5.

5.2. Evaluation metrics

In this study, to measure the performance of the proposed threat detection system, we use Loss, Detection Rate (DR...), and Area Under the Curve (AUC). These metrics have been widely used in the literature (e.g. Loss [38,35,59], DR... [58,37,45], AUC [60,61,40]). Detection Rate (DR...), also called True Positive Rate (TPR), is defined as follows:

$$TPR = \frac{True\ Positives}{True\ Positive + False\ Negatives} \quad (4)$$

And False Positive Rate (FPR) is defined as:

$$FPR = \frac{False\ Positives}{False\ Positives + True\ Negatives} \quad (5)$$

Furthermore, Receiving Characteristic Curve (ROC) is a curve that describes the relationship between TPR and FPR under different decision thresholds, and AUC (Area Under the Curve), which is the area under the ROC curve, is the measure of the performance across all possible thresholds. Moreover, Loss is a metric to calculate the distance between the real and predicted values. In this paper, we utilize Mean Squared Error (MSE) for loss calculation.

5.3. Experimental setup

We designed some experiments to evaluate the performance of the proposed threat detection system. The proposed learning model is developed in TensorFlow. In each training round, each participant uses local data to train a local autoencoder model. To find the best configuration for the autoencoder model, we utilized the Grid Search approach. The used autoencoder has 5 layers. The number of neurons in the first and the last layers are equal to the number of features, the second and the third layer has 443 neurons and the middle layer consists of 221 neurons. The *relu* activation function with a learning rate equal to 0.001 is utilized in the designed autoencoder. The initial weights of the models are set randomly and 11 different local participants collaborate in the learning process. For evaluating explainability, we used *KernelExplainer* from the SHAP library and considered the top 5 anomalous features from the sample with the highest reconstruction error. To study the impact of different features on the anomalous features, we remove the anomalous feature weights from the autoencoder and use the *KernelExplainer* to calculate the Shapely values for other features.

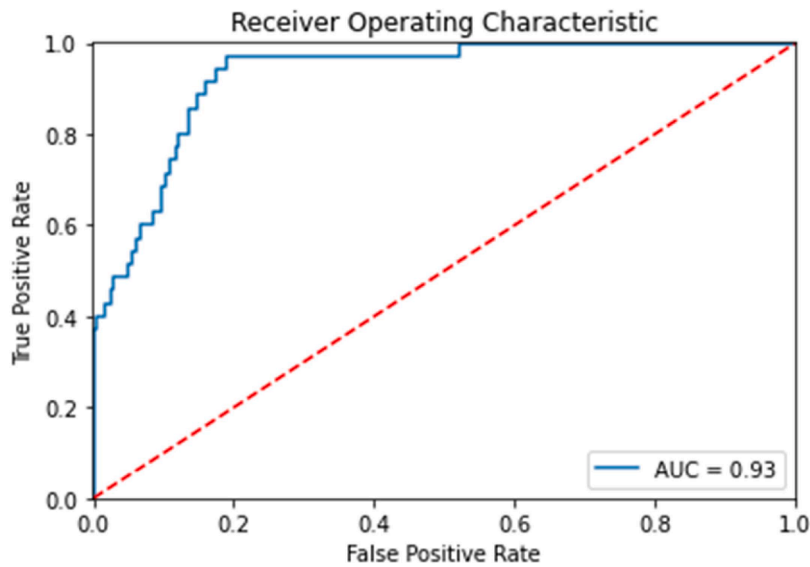


Fig. 4. ROC and AUC of the Presented System.

Table 7

Performance Comparison with Centralized Solutions.

Paper	AUC
Matterer and LeJeune [40]	0.81
Liu et al. [41]	0.93
Le and Zincir-Heywood [60]	0.95
The Proposed Solution	0.93

5.4. Experimental results

5.4.1. Model performance

This section reports the results of applying the proposed method for insider threat detection in IoT. To this end, first, we examined the performance of the local models in the federated learning process. Then we analyzed the final global model and compared it with other proposed approaches in the literature under different amounts of investigation budgets. The investigation budget refers to the portion of actions that are selected to be investigated as a possible threat.

The Losses of these models in the federated learning procedure are illustrated in Table 6. The first row in the table shows the loss values for different participants when they aim to individually detect insider threats. The remaining rows demonstrate the loss value in the federated learning in which participants utilize the global model generated in the previous round as the initial model. The experiment was repeated in 10 rounds. As can be seen in the table, the average value for loss value increased in the first 5 rounds and after that, we can see no considerable changes in the loss values. Although in some participants (e.g. participant 1) the collaboration with others doesn't lead to a big improvement in the model's performance, in most participants federated learning enhances the performance. It can be seen in Table 6 that the federated learning approach outperforms the individual learning in which every participant tries to train a model independently. Federated learning provides an opportunity for all participants to use the cooperation of other participants without getting direct access to their local data.

The curve in Fig. 4 represents the relation between the TPR and the FPR of the proposed model for different thresholds for suspicious behavior detection. The higher the value of the AUC score means the better performance in threat detection. The obtained AUC value in the proposed system is 93%. Moreover, compared to the central solutions provided in the literature, the proposed system reaches promising performance in terms of AUC. The AUC values for other studies that worked on CERT v6.2 are shown in Table 7. As can be seen in this table, the proposed solution reaches a high AUC value close to the best-proposed solutions in the literature, while all other solutions are central and provide no mechanism for users' data privacy preservation. In the central solutions, the participants have been assumed to be open to sharing their data with the central server which is not a reasonable assumption because the data that is utilized for human behavior analysis is usually personal and confidential. Considering this fact, we highly recommend the proposed approach in this study which is capable of detecting most of the threats in the system using a limited investigation budget while preserving the privacy of user's data.

Another metric that has been utilized in the literature to demonstrate the performance of insider threat detection solutions is the investigation budget. The investigation budget represents the portion of actions that have been selected for further investigation and

Table 8
Detection Rate Under Different Investigation Budgets.

Investigation Budget	0.1%	1%	5%	10%	15%	20%
Le and Zincir-Heywood [60]	0.26	0.39	0.67	0.84	–	0.96
The Proposed Solution	0.34	0.4	0.51	0.69	0.89	0.97

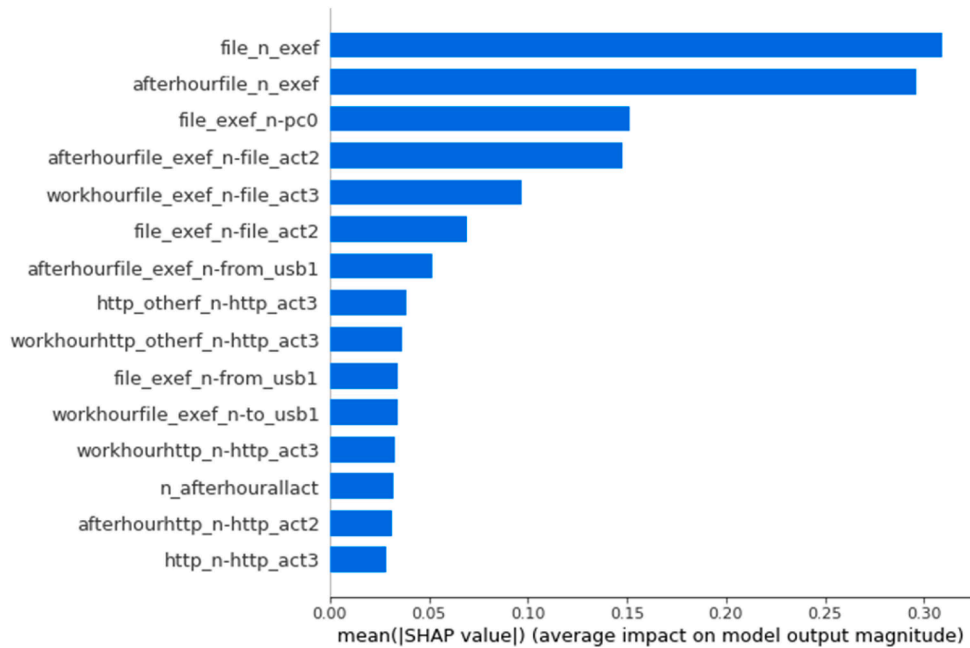


Fig. 5. Average Contribution in Anomalous Features.

the percentage of the detected threats. From the other studies that worked on CERT v6.2, Le and Zincir-Heywood [60] have reported this metric. Table 8 compares the Detection Rate of the proposed model with the results provided by Le and Zincir-Heywood. [60] As can be seen in this table, both models achieve close performance while the solution presented by Le and Zincir-Heywood [60] has the privilege of having access to all data samples centrally. As can be seen in this table, the proposed approach can detect 97% of the threats by investigating only 20% of the most anomalous instances.

5.4.2. Explainability

As described in Section 3, the proposed approach is capable of explaining why a sample is detected as an anomaly. To this aim, we seek the samples with the highest reconstruction error and find the features that are mostly responsible for this error. We call these features anomalous features and our final goal is to assess the contribution of all features in making these features anomalous. The experiments show that the top 5 anomalous features in our dataset are *afterhourhttp_mean_url_len*, *afterhourhttp_otherf_mean_url_depth*, *afterhourhttp_otherf_mean_url_len*, *afterhourhttp_otherf_mean_http_c_nwords*, *file_mean_file_nwords*. Fig. 5 shows the top 15 features that have the most average contribution in the anomaly score of these anomalous features and Fig. 6 demonstrates the most contributing features in each anomalous feature separately.

Fig. 5 shows that the number of file executions in a day on average has the most impact on the anomalous features. In addition to this feature, most of the other features are related to executing a file on a shared computer or after work hours. This explanation sheds light on the behaviors that might cause a malicious action. By monitoring these features, suspicious actions can be detected to be considered for further security analysis.

In Fig. 6 features' contributions to anomalous features are depicted separately using the waterfall plot. The waterfall plot demonstrates the positive and negative role of each feature in the output results. The red features have a negative impact and cause anomalies, while the blue features are the ones that have a positive impact and try to prevent an anomaly. A waterfall plot also shows the amount of contribution for contributing features. As is shown in Fig. 6(a), the most negative contributing features in *afterhourhttp_mean_url_len* is the number of executed files in the work hours (*file_n_exef*) and the number of executed files after work hours (*afterhourfile_n_exef*) that make 54% contribution in total, while the number of files that are used just for read is a feature that decreases the chance for being anomalous by 22%. For *afterhourhttp_otherf_mean_url_depth* (Fig. 6(b)), *afterhourhttp_otherf_mean_url_len* (Fig. 6(c)), and *afterhourhttp_otherf_mean_http_c_nwords* (Fig. 6(d)), we have the same pattern, and the two most contributing positive features and also the negative feature are the same. It is different in *file_mean_file_nwords* (Fig. 6(e)), where the contributing features are different



Fig. 6. Waterfall Charts for the top 5 Anomalous Features: (a) for afterhourhttp_mean_url_len, (b) for fterhourhttp_otherf_mean_url_depth, (c) for afterhourhttp_otherf_mean_url_len, (d) for afterhourhttp_otherf_mean_http_c_nwords, and (e) for file_mean.

from the other anomalous features, and also the contribution values for all the features are less than 6%.

6. Discussion

In this paper, we showed that the proposed solution provides a privacy-preserving and explainable approach for insider threat detection in IoT. The proposed solution also outperforms the local models in the federated context. The reason for this superior performance is that the models trained over local data in the federated model, although they do not have access to all training sets (Table 6), obtain an overall insight into data through the parameters of the locally trained models. This insight enables the global model to get a more comprehensive knowledge of the potential insider threats that might occur in the system. Moreover, similar to other approaches that use an unsupervised approach for threat detection, the proposed approach can be utilized to find early signs of suspicious actions and report these actions to cybersecurity experts for further investigation. Cybersecurity experts can benefit from the proposed solution in two aspects. First, the presented approach can detect most insider threats under very low investigation

budgets. Second, the provided explanation of the most impacting features in user behavior helps the experts to take action to prevent such malicious behaviors and safeguard the system against insider threats.

Furthermore, the average contribution of different users' actions in insider threats shows that many of the malicious actions are correlated with the number of file executions and also the time that the file execution happens. Another major factor in malicious actions is logging into untrusted websites for uploading or downloading files. As a result, it is vital to use updated antivirus and firewall tools along with continuous network monitoring for threat detection.

Finally, it is worth mentioning that currently an issue in research on insider threat detection in IoT is the limited availability of benchmark data. Many of the available public datasets on user behavior are not generated in IoT environments. Although some of these datasets can be utilized for studying IoT insider threat detection (Kin et al. [39]), data sources that specifically are extracted from IoT users can facilitate studies in this area and allow researchers to study the different aspects of user interactions in IoT environments.

7. Conclusion

Insider attacks, where authorized users intentionally or unintentionally pose threats from inside, are a significant concern in IoT. While machine learning (ML) has been widely employed in cybersecurity, its use for insider threat detection in IoT has been limited due to privacy concerns about sharing user behavior data with others. To tackle this problem, in this paper, we proposed a privacy-preserving approach for insider threat detection in IoT. The presented solution is capable of training the local models on the clients' side therefore the users' data remains protected. Another issue in the threat detection systems that utilize deep learning is they work as a black box which means that they can provide a decision but are not capable of explaining the reasons for coming up with the decision. This study introduces a model that offers explainable decisions, mitigating this issue in threat detection. The results demonstrate a 93% AUC value which is close to the performance of similar central solutions while eliminating the privacy concerns about data sharing. However, a significant challenge in this study, reflective of a broader issue in IoT insider threat detection research, is the limited availability of benchmark data which are generated in real IoT environments. Future directions may involve investigating alternative datasets and developing federated models to analyze user actions as time series data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This research was funded by a Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant awarded to R. Dara.

References

- [1] S.M. Tahsien, H. Karimipour, P. Spachos, Machine learning based solutions for security of Internet of Things (IoT): a survey, *J. Netw. Comput. Appl.* (2020) 161, <https://doi.org/10.1016/j.jnca.2020.102630>.
- [2] M. Amiri-Zarandi, R.A. Dara, E. Fraser, LBTM: a lightweight blockchain-based trust management system for social internet of things, *J. Supercomput.* (2022), <https://doi.org/10.1007/s11227-021-04231-3>.
- [3] M. Amiri-Zarandi, M. Hazrati Fard, S. Yousefinaghani, M. Kaviani, R. Dara, A platform approach to smart farm information processing, *Agriculture* 12 (2022) 838, <https://doi.org/10.3390/agriculture12060838>.
- [4] R. Mahmoud, T. Yousuf, F. Aloul, I. Zualkernan, Internet of things (IoT) security: current status, challenges and prospective measures, *Int. Conf. Internet Technol. Secur. Trans. ICITST.* (2015) 336–341, <https://doi.org/10.1109/ICITST.2015.7412116>.
- [5] H.M. Rouzbahani, Z. Faraji, M. Amiri-Zarandi, H. Karimipour, AI-enabled Security Monitoring in Smart Cyber Physical Grids, in: *Secure Cyber-Physical System*, Springer, 2020, pp. 145–167.
- [6] M. Amiri-Zarandi, R.A. Dara, E. Fraser, A survey of machine learning-based solutions to protect privacy in the Internet of Things, *Comput. Secur.* 96 (2020), 101921, <https://doi.org/10.1016/j.cose.2020.101921>.
- [7] A. Yazdinejad, A. Dehghantanha, R.M. Parizi, G. Srivastava, H. Karimipour, Secure intelligent fuzzy blockchain framework: effective threat detection in iot networks, *Comput. Ind.* 144 (2023), 103801, <https://doi.org/10.1016/j.compind.2022.103801>.
- [8] A.N. Jahromi, H. Karimipour, A. Dehghantanha, K.K.R. Choo, Toward detection and attribution of cyber-attacks in IOT-enabled cyber-physical systems, *IEEE Internet Things J.* 8 (2021) 13712–13722, <https://doi.org/10.1109/JIOT.2021.3067667>.
- [9] M. Amiri-Zarandi, R.A. Dara. Blockchain-based Trust Management in Social Internet of Things., *IEEE Conference Internet of things (IoT) security: Current status, challenges and prospective measuresI*, 2020, pp. 49–54, <https://doi.org/10.1109/dasc-picom-cybercitech49142.2020.00024>.
- [10] A.Y. Khan, R. Latif, S. Latif, S. Tahir, G. Batool, T. Saba, Malicious insider attack detection in IoTs using data analytics, *IEEE Access* 8 (2020) 11743–11753, <https://doi.org/10.1109/ACCESS.2019.2959047>.
- [11] H. Schulze, 2020 Insider Threat Survey Report. Accessed: Jan. 29, 2020, <https://gurucul.com/2020-insider-threat-survey-report>.
- [12] L. Tawalbeh, F. Muheidat, M. Tawalbeh, M. Quwaider, IoT privacy and security: challenges and solutions, *Appl. Sci.* 10 (2020) 1–17, <https://doi.org/10.3390/AP10124102>.

- [13] Z. Al-Mhiqani, Mohammed Nasser, Rabiah Ahmad, Z. Zainal Abidin, Warusia Yassin, Aslinda Hassan, Karrar Hameed Abdulkareem, Ali Nabeel Salih, Yunos, A review of insider threat detection Classification, machine learning techniques, datasets, open challenges, and recommendations, *Appl. Sci.* 10 (2022) 5208.
- [14] M. Amiri-Zarandi, A Privacy-Preserving Trust Management Framework For IoT, University of Guelph, 2022.
- [15] K. Siau, W. Wang, Building trust in artificial intelligence, machine learning, and robotics, *Cut. Bus. Technol. J.* 31 (2018) 47–53.
- [16] S. Majumdar, Fairness, explainability, privacy, and robustness for trustworthy algorithmic decision making, *Big Data Analytics in Chemoinformatics and Bioinformatics*, 2023, pp. 61–95.
- [17] D. Franco, L. Oneto, N. Navarin, D. Anguita, Toward learning trustworthily from data combining privacy, fairness, and explainability: an application to face recognition, *Entropy* 23 (2021), <https://doi.org/10.3390/e23081047>.
- [18] E. Toreini, M. Aitken, K. Coopamotto, K. Elliott, C.G. Zelaya, A. van Moorsel, The relationship between trust in AI and trustworthy machine learning technologies, in: *Proceedings of the FAT* Conference Fairness, Accountability, Transparency*, 2020, pp. 272–283, <https://doi.org/10.1145/3351095.3372834>.
- [19] A. Kumar, T. Braud, S. Tarkoma, P. Hui, Trustworthy ai in the age of pervasive computing and big data, *Proceedings of the IEEE International Conference Pervasive Computation Communication Work PerCom Work* 2020. (2020). 10.1109/PerComWorkshops48775.2020.9156127.
- [20] R. Shirey, Internet Security Glossary, Version 2, RFC 4949. (2007). 10.17487/rfc4949.
- [21] E. Bertino, E. Ferrari, Big data security and privacy, *Stud. Big Data* 31 (2018) 425–439, https://doi.org/10.1007/978-3-319-61893-7_25.
- [22] M. Amiri-Zarandi, R.A. Dara, E. Duncan, E. Fraser, Big data privacy in smart farming: a review, *Sustainability* 14 (2022) 9120.
- [23] J. Kaur, S.M. Hazrati Fard, M. Amiri-Zarandi, R. Dara, Protecting farmers' data privacy and confidentiality: recommendations and considerations, *Front. Sustain. Food Syst.* (n.d.) 2023 475.
- [24] A. Rai, Explainable AI: from black box to glass box, *J. Acad. Mark. Sci.* 48 (2020) 137–141, <https://doi.org/10.1007/s11747-019-00710-5>.
- [25] J. Konecny, H.B. McMahan, F.X. Yu, P. Richtárik, A.T. Suresh, D. Bacon, Federated learning: strategies for improving communication efficiency, *ArXiv preprint*. (2016) ArXiv1610.05492. <http://arxiv.org/abs/1610.05492>.
- [26] M. Ali, H. Karimipour, M. Tariq, Integration of blockchain and federated learning for Internet of Things: recent advances and future challenges, *Comput. Secur.* 108 (2021), 102355, <https://doi.org/10.1016/j.cose.2021.102355>.
- [27] A.N. Jahromi, H.K. Schulich, A. Dehghantanha, Deep federated learning-based cyber-attack detection in industrial control systems, 18th Int. Conf. Privacy, Secur. Trust. PST. (2021), <https://doi.org/10.1109/PST52912.2021.9647838>.
- [28] M. Amiri-Zarandi, R.A. Dara, X. Lin, SIDS: a federated learning approach for intrusion detection in IoT using Social Internet of Things, *Comput. Netw.* 236 (2023), 110005, <https://doi.org/10.1016/j.comnet.2023.110005>.
- [29] A. Yazdinejad, A. Dehghantanha, R.M. Parizi, M. Hammoudeh, H. Karimipour, G. Srivastava, Block Hunter, Federated learning for cyber threat hunting in blockchain-based IIoT networks, *IEEE Trans. Ind. Inf.* (2022) 1–10, <https://doi.org/10.1109/TII.2022.3168011>.
- [30] Y. Zhao, S. Member, J. Zhao, Mobile edge computing, blockchain and reputation-based crowdsourcing iot federated learning : a secure Decentralized and Privacy-preserving System, *arXiv preprint* (2019) 2327–4662.
- [31] D. Doran, S. Schulz, T.R. Besold, What does explainable AI really mean? A new conceptualization of perspectives, *arXiv preprint* (2017) arXiv:1710.00794.
- [32] A. Pakula, From local explanations to global understanding with explainable AI for trees, *Methods Mol. Biol.* 176 (2019) 139–148, <https://doi.org/10.1038/s42256-019-0138-9>.
- [33] E. Lee, D. Braines, M. Stiffler, A.A. Hudler, D. Harborne, Developing the sensitivity of LIME for better machine learning explanation, *SPIE* (2019), <https://doi.org/10.1117/12.2520149>.
- [34] N. Takeishi, Shapley values of reconstruction errors of PCA for explaining anomaly detection, *IEEE Int. Conf. Data Min. Work. ICDMW.* (2019) 793–798, <https://doi.org/10.1109/ICDMW.2019.00117>.
- [35] J. Lu, R.K. Wong, Insider threat detection with long short-term memory, *ACM Int. Conf. Proceeding Ser.* (2019), <https://doi.org/10.1145/3290688.3290692>.
- [36] M.N. Al-Mhiqani, R. Ahmad, Z.Z. Abidin, W. Yassin, A. Hassan, A.N. Mohammad, New insider threat detection method based on recurrent neural networks, *Indones. J. Electr. Eng. Comput. Sci.* 17 (2019) 1474–1479, <https://doi.org/10.11591/ijeecs.v17.i3.pp1474-1479>.
- [37] B. Sharma, P. Pokharel, B. Joshi, User behavior analytics for anomaly detection using LSTM autoencoder-insider threat detection, *ACM Int. Conf. Proceeding Ser.* (2020), <https://doi.org/10.1145/3406601.3406610>.
- [38] L. Liu, O. De Vel, C. Chen, J. Zhang, Y. Xiang, Anomaly-based insider threat detection using deep autoencoders, *IEEE Int. Conf. Data Min. Work. ICDMW.* (2019) 39–48, <https://doi.org/10.1109/ICDMW.2018.00014>.
- [39] A. Kim, J. Oh, J. Ryu, K. Lee, A review of insider threat detection approaches with IoT perspective, *IEEE Access* 8 (2020) 78847–78867, <https://doi.org/10.1109/ACCESS.2020.2990195>.
- [40] J. Matterer, D. Lejeune, Peer group metadata-informed LSTM ensembles for insider threat detection, *Proc. 31st Int. Florida Artif. Intell. Res. Soc. Conf.* (2018) 62–67.
- [41] F. Liu, X. Jiang, Y. Wen, X. Xing, D. Zhang, D. Meng, Log2vec: a heterogeneous graph embedding based approach for detecting cyber threats within enterprise, *Proc. ACM Conf. Comput. Commun. Secur.* (2019) 1777–1794, <https://doi.org/10.1145/3319535.3363224>.
- [42] S. Yuan, X. Wu, Deep learning for insider threat detection: review, challenges and opportunities, *Comput. Secur.* 104 (2021), 102221, <https://doi.org/10.1016/j.cose.2021.102221>.
- [43] S.A. Rahman, H. Tout, C. Talhi, A. Mourad, Internet of things intrusion detection: centralized, on-device, or federated learning? *IEEE Netw.* 34 (2020) 310–317, <https://doi.org/10.1109/MNET.011.2000286>.
- [44] M. Sarhan, S. Layeghy, N. Moustafa, M. Portmann, Cyber threat intelligence sharing scheme based on federated learning for network intrusion detection, *J. Netw. Syst. Manag.* 31 (1) (2021).
- [45] V. Rey, P.M.S. Sánchez, A.H. Celdrán, G. Bovet, M. Jaggi, Federated learning for malware detection in IoT devices, *Comput. Netw.* 204 (2022), 108693. <http://arxiv.org/abs/2104.09994>.
- [46] D. Preuveneers, V. Rimmer, I. Tsiengenopoulos, J. Spooren, W. Joosen, E. Ilie-Zudor, Chained anomaly detection models for federated learning: an intrusion detection case study, *Appl. Sci.* 8 (2018) 1–21, <https://doi.org/10.3390/app8122663>.
- [47] Y. Fan, Y. Li, M. Zhan, H. Cui, Y. Zhang, IoTDefender: a federated transfer learning intrusion detection framework for 5G IoT, *Proc. - 2020 IEEE 14th Int. Conf. Big Data Sci. Eng. BigDataSE 2020* (2020) 88–95.
- [48] T.W. Kim, B.R. Routledge, Informational privacy, a right to explanation, and interpretable AI, *2023 Proc. - 2018 2nd IEEE Symp. Privacy-Aware Comput. PAC 2018* (2018) 64–74.
- [49] M. Kopp, T. Pevný, M. Holeňa, Anomaly explanation with random forests, *Expert Syst. Appl.* (2020) 149, <https://doi.org/10.1016/j.eswa.2020.113187>.
- [50] L. Antwarg, R.M. Miller, B. Shapira, L. Rokach, Explaining Anomalies Detected by Autoencoders Using SHAP, *ArXiv Prepr* (2019) 1–37. <http://arxiv.org/abs/1903.02407>. ArXiv1903.02407.
- [51] A. Azaria, A. Richardson, S. Kraus, V.S. Subrahmanian, Behavioral analysis of insider threat: a survey and bootstrapped prediction in imbalanced data, *ArXiv Prepr.* (2014) 135–155. ArXiv1901.03407.
- [52] A. Al-Abassi, J. Sakhnini, H. Karimipour, Unsupervised stacked autoencoders for anomaly detection on smart cyber-physical grids, *Conf. Proc. - IEEE Int. Conf. Syst. Man Cybern.* (2020) 3123–3129, <https://doi.org/10.1109/SMC42975.2020.9283064>.
- [53] R. Chalaphathy, S. Chawla, Deep learning for anomaly detection: a survey, *ArXiv Prepr* (2019) 1–50. <http://arxiv.org/abs/1901.03407>. ArXiv1901.03407.
- [54] H. Mohammadi Rouzbahani, H. Karimipour, A. Rahimnejad, A. Dehghantanha, G. Srivastava, Anomaly Detection in Cyber-Physical Systems Using Machine learning, in: *Handbook Big Data Privacy*, Springer, 2020, pp. 219–235.
- [55] G. Pang, C. Shen, L. Cao, A. Van Den Hengel, Deep learning for anomaly detection: a review, *ACM Comput. Surv.* (2021) 54, <https://doi.org/10.1145/3439950>.
- [56] H. Choi, M. Kim, G. Lee, W. Kim, Unsupervised learning approach for network intrusion detection system using autoencoders, *J. Supercomput.* 75 (2019) 5597–5621, <https://doi.org/10.1007/s11227-019-02805-w>.
- [57] J. Glasser, B. Lindauer, Bridging the gap: a pragmatic approach to generating insider threat data, *Proc. - IEEE CS Secur. Priv. Work. SPW* (2013). (2013) 98–104, <https://doi.org/10.1109/SPW.2013.37>.

- [58] D.C. Le, N. Zincir-Heywood, M.I. Heywood, Analyzing data granularity levels for insider threat detection using machine learning, *IEEE Trans. Netw. Serv. Manag.* 17 (2020) 30–44, <https://doi.org/10.1109/TNSM.2020.2967721>.
- [59] A. Tuor, S. Kaplan, B. Hutchinson, N. Nichols, S. Robinson, Deep learning for unsupervised insider threat detection in structured cybersecurity data streams, *AAAI Work. - Tech. Rep.* (2017) 224–234.
- [60] D.C. Le, N. Zincir-Heywood, Anomaly detection for insider threats using unsupervised ensembles, *IEEE Trans. Netw. Serv. Manag.* 18 (2021) 1152–1164, <https://doi.org/10.1109/TNSM.2021.3071928>.
- [61] F. Yuan, Y. Cao, Y. Shang, Y. Liu, Insider threat detection with deep neural network, *18th Conf. on Computational Science-ICCS (2018)* 43–54, <https://doi.org/10.1007/978-3-319-93698-7>.